Lab 4 - CNN Report

COS30082 Applied Machine Learning

Truong Nam Hung - 105556375

Dataset: 6 classes (buildings, forest, glacier, mountain, sea, street). seg_train was split 80% training / 20% validation; seg_test was used only for final evaluation. Images were resized to 150×150 and normalized to [0, 1].

1. Model architectures

Input is 150×150×3 for all models. All Conv2D layers use 3×3 kernels with ReLU; all MaxPooling layers are 2×2.

Model	Architecture	Parameters
Task 2: Baseline	Conv(32) → MaxPool → Flatten → Dense(128) → Dense(6, softmax)	22,431,494
Task 3: Modified	[Conv(32) → MaxPool] → [Conv(64) → MaxPool] → [Conv(128) → MaxPool] → Flatten → Dense(128) → Dense(6, softmax)	4,829,126
Task 4: Augmented	Augmentation layers (flip, rotation, zoom) → same as Task 3	4,829,126
Task 5: Regularized	Same as Task 3, with Dropout(0.5) after Dense(128)	4,829,126

The extra pooling in the modified model shrinks the flattened vector from 175,232 to 36,992 values, cutting the parameter count by about 80%.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 148, 148, 32)	896
max_pooling2d (MaxPooling2D)	(None, 74, 74, 32)	0
flatten (Flatten)	(None, 175232)	0
dense (Dense)	(None, 128)	22,429,824
dense_1 (Dense)	(None, 6)	774

Total params: 22,431,494 (85.57 MB)
Trainable params: 22,431,494 (85.57 MB)
Non-trainable params: 0 (0.00 B)

Figure 1. Baseline CNN

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 148, 148, 32)	896
max_pooling2d_1 (MaxPooling2D)	(None, 74, 74, 32)	0
conv2d_2 (Conv2D)	(None, 72, 72, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 36, 36, 64)	0
conv2d_3 (Conv2D)	(None, 34, 34, 128)	73,856
max_pooling2d_3 (MaxPooling2D)	(None, 17, 17, 128)	0
flatten_1 (Flatten)	(None, 36992)	0
dense_2 (Dense)	(None, 128)	4,735,104
dense_3 (Dense)	(None, 6)	774

Total params: 4,829,126 (18.42 MB)
Trainable params: 4,829,126 (18.42 MB)
Non-trainable params: 0 (0.00 B)

Figure 2. Modified CNN architecture

Augmented CNN created successfully.
Model: "augmented_cnn"

Layer (type)	Output Shape	Param #
data_augmentation (Sequential)	(None, 150, 150, 3)	0
conv2d_4 (Conv2D)	(None, 148, 148, 32)	896
max_pooling2d_4 (MaxPooling2D)	(None, 74, 74, 32)	0
conv2d_5 (Conv2D)	(None, 72, 72, 64)	18,496
max_pooling2d_5 (MaxPooling2D)	(None, 36, 36, 64)	0
conv2d_6 (Conv2D)	(None, 34, 34, 128)	73,856
max_pooling2d_6 (MaxPooling2D)	(None, 17, 17, 128)	0
flatten_2 (Flatten)	(None, 36992)	0
dense_4 (Dense)	(None, 128)	4,735,104
dense_5 (Dense)	(None, 6)	774

Total params: 4,829,126 (18.42 MB)
Trainable params: 4,829,126 (18.42 MB)
Non-trainable params: 0 (0.00 B)

Figure 3. Augmented CNN

Regularized CNN created successfully.
Model: "regularized_cnn"

Layer (type)	Output Shape	Param #
conv2d_7 (Conv2D)	(None, 148, 148, 32)	896
max_pooling2d_7 (MaxPooling2D)	(None, 74, 74, 32)	0
conv2d_8 (Conv2D)	(None, 72, 72, 64)	18,496
max_pooling2d_8 (MaxPooling2D)	(None, 36, 36, 64)	0
conv2d_9 (Conv2D)	(None, 34, 34, 128)	73,856
max_pooling2d_9 (MaxPooling2D)	(None, 17, 17, 128)	0
flatten_3 (Flatten)	(None, 36992)	0
dense_6 (Dense)	(None, 128)	4,735,104
dropout (Dropout)	(None, 128)	0
dense_7 (Dense)	(None, 6)	774

Total params: 4,829,126 (18.42 MB)
Trainable params: 4,829,126 (18.42 MB)
Non-trainable params: 0 (0.00 B)

Figure 4. Regularised CNN

2. Training settings

- **Optimizer:** Adam (default learning rate 0.001); **Loss:** sparse categorical cross-entropy; **Metric:** accuracy
- **Batch size:** 32; **Random seed:** 42
- **Epochs:** 10 for Tasks 2–4. Task 5 had a maximum of 15; EarlyStopping ended it after 7.

3. Data augmentation (Task 4)

Applied to training images only (switched off at validation and test time): random horizontal flip, random rotation (up to ±18°), and random zoom (up to 10%).

4. Regularization (Task 5)

Dropout (rate 0.5) after the hidden dense layer, and **EarlyStopping** monitoring validation loss (patience 3, best weights restored).

5. Test accuracy comparison

Task	Model	Val. accuracy	Test accuracy	Test loss	Epochs run
2	Baseline CNN	70.17%	71.23%	1.7902	10
3	Modified CNN	79.58%	78.57%	1.1831	10
4	+ Data augmentation	76.59%	77.47%	0.8086	10
5	+ Dropout & EarlyStopping	80.72%	**81.37%**	**0.5532**	**7**
Extra	Augmentation + Dropout + EarlyStopping	83.39%	83.40%	0.5159	17

Note: The extra row is not part of Tasks 2 to 5.

Discussion: The regularized CNN (Task 5) scored highest among Tasks 2 to 5 at 81.37%. The deeper architecture in Task 3 raised test accuracy by about 7 points over the baseline (71.23% → 78.57%) while using far fewer parameters; the baseline's high test loss (1.79) suggests it overfit. Augmentation (Task 4) did not raise test accuracy (77.47% vs 78.57%) but cut test loss from 1.18 to 0.81, suggesting less overfitting; with only 10 epochs it may not have finished learning, as augmented training converges more

slowly. Dropout with EarlyStopping gave the best result, stopping after 7 epochs. Combining all three techniques reached 83.40% test accuracy, showing the techniques work well together.